

DATA 505 (Statistics Using R): Exam 1

Part 2: Open Computer

Name: Sumyyah Kadam

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze data about Antarctic penguins from the Palmer Station research program. The data are available in the `palmerpenguins` R package.

```
# Install and load the required packages
if(!"palmerpenguins" %in% installed.packages()){
  install.packages("palmerpenguins")
}
if(!"ggplot2" %in% installed.packages()){
  install.packages("ggplot2")
}

library(palmerpenguins)
library(ggplot2)

# Load the penguins data
data("penguins")
```

Data Exploration (4 points)

1 (2 points)

Use the `str()` function to examine the structure of the penguins dataset.

```
str(penguins)
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm  : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

How many observations and variables are there?

Answer: There are 344 observations and 8 variables.

2 (2 points)

Extract the `bill_length_mm` column from the penguins dataset and store it in a new variable called `bill_lengths`. Then use the `summary()` function to examine this variable.

```
bill_lengths = penguins$bill_length_mm;bill_lengths
```

```
[1] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42.0 37.8 37.8 41.1 38.6 34.6
[16] 36.6 38.7 42.5 34.4 46.0 37.8 37.7 35.9 38.2 38.8 35.3 40.6 40.5 37.9 40.5
[31] 39.5 37.2 39.5 40.9 36.4 39.2 38.8 42.2 37.6 39.8 36.5 40.8 36.0 44.1 37.0
[46] 39.6 41.1 37.5 36.0 42.3 39.6 40.1 35.0 42.0 34.5 41.4 39.0 40.6 36.5 37.6
[61] 35.7 41.3 37.6 41.1 36.4 41.6 35.5 41.1 35.9 41.8 33.5 39.7 39.6 45.8 35.5
[76] 42.8 40.9 37.2 36.2 42.1 34.6 42.9 36.7 35.1 37.3 41.3 36.3 36.9 38.3 38.9
[91] 35.7 41.1 34.0 39.6 36.2 40.8 38.1 40.3 33.1 43.2 35.0 41.0 37.7 37.8 37.9
[106] 39.7 38.6 38.2 38.1 43.2 38.1 45.6 39.7 42.2 39.6 42.7 38.6 37.3 35.7 41.1
[121] 36.2 37.7 40.2 41.4 35.2 40.6 38.8 41.5 39.0 44.1 38.5 43.1 36.8 37.5 38.1
[136] 41.1 35.6 40.2 37.0 39.7 40.2 40.6 32.1 40.7 37.3 39.0 39.2 36.6 36.0 37.8
```

```
[151] 36.0 41.5 46.1 50.0 48.7 50.0 47.6 46.5 45.4 46.7 43.3 46.8 40.9 49.0 45.5
[166] 48.4 45.8 49.3 42.0 49.2 46.2 48.7 50.2 45.1 46.5 46.3 42.9 46.1 44.5 47.8
[181] 48.2 50.0 47.3 42.8 45.1 59.6 49.1 48.4 42.6 44.4 44.0 48.7 42.7 49.6 45.3
[196] 49.6 50.5 43.6 45.5 50.5 44.9 45.2 46.6 48.5 45.1 50.1 46.5 45.0 43.8 45.5
[211] 43.2 50.4 45.3 46.2 45.7 54.3 45.8 49.8 46.2 49.5 43.5 50.7 47.7 46.4 48.2
[226] 46.5 46.4 48.6 47.5 51.1 45.2 45.2 49.1 52.5 47.4 50.0 44.9 50.8 43.4 51.3
[241] 47.5 52.1 47.5 52.2 45.5 49.5 44.5 50.8 49.4 46.9 48.4 51.1 48.5 55.9 47.2
[256] 49.1 47.3 46.8 41.7 53.4 43.3 48.1 50.5 49.8 43.5 51.5 46.2 55.1 44.5 48.8
[271] 47.2    NA 46.8 50.4 45.2 49.9 46.5 50.0 51.3 45.4 52.7 45.2 46.1 51.3 46.0
[286] 51.3 46.6 51.7 47.0 52.0 45.9 50.5 50.3 58.0 46.4 49.2 42.4 48.5 43.2 50.6
[301] 46.7 52.0 50.5 49.5 46.4 52.8 40.9 54.2 42.5 51.0 49.7 47.5 47.6 52.0 46.9
[316] 53.5 49.0 46.2 50.9 45.5 50.9 50.8 50.1 49.0 51.5 49.8 48.1 51.4 45.7 50.7
[331] 42.5 52.2 45.2 49.3 50.2 45.6 51.9 46.8 45.7 55.8 43.5 49.6 50.8 50.2
```

```
summary(bill_lengths)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
32.10	39.23	44.45	43.92	48.50	59.60	2

Research Question

We want to investigate whether the mean bill length of Adelie penguins differs significantly from 39 mm. This research question is motivated by previous studies suggesting that 39 mm might be a typical bill length for this species.

1 (4 points)

Filter the dataset to include only Adelie penguins and store the result in a new variable called `adelie_penguins`. Remove any rows with missing bill length measurements.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v lubridate  1.9.4      v tibble     3.2.1
v purrr      1.1.0      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()      masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
adelie_penguins = penguins |>
  filter(penguins$species=="Adelie") |>
  na.omit(penguins$bill_length_mm)
adelie_penguins
```

```
# A tibble: 146 x 8
  species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
  <fct>   <fct>         <dbl>         <dbl>             <int>         <int>
1 Adelie Torgersen      39.1           18.7             181          3750
2 Adelie Torgersen      39.5           17.4             186          3800
3 Adelie Torgersen      40.3           18              195          3250
4 Adelie Torgersen      36.7           19.3             193          3450
5 Adelie Torgersen      39.3           20.6             190          3650
6 Adelie Torgersen      38.9           17.8             181          3625
7 Adelie Torgersen      39.2           19.6             195          4675
8 Adelie Torgersen      41.1           17.6             182          3200
9 Adelie Torgersen      38.6           21.2             191          3800
10 Adelie Torgersen      34.6           21.1             198          4400
# i 136 more rows
# i 2 more variables: sex <fct>, year <int>
```

2 (3 points)

Extract the bill lengths from the Adelie penguin data and store them in a variable called `adelie_bills`. How many Adelie penguins have complete bill length measurements?

```
adelie_bills = as.numeric(adelie_penguins[["bill_length_mm"]])
adelie_bills
```

```
[1] 39.1 39.5 40.3 36.7 39.3 38.9 39.2 41.1 38.6 34.6 36.6 38.7 42.5 34.4 46.0
[16] 37.8 37.7 35.9 38.2 38.8 35.3 40.6 40.5 37.9 40.5 39.5 37.2 39.5 40.9 36.4
[31] 39.2 38.8 42.2 37.6 39.8 36.5 40.8 36.0 44.1 37.0 39.6 41.1 36.0 42.3 39.6
[46] 40.1 35.0 42.0 34.5 41.4 39.0 40.6 36.5 37.6 35.7 41.3 37.6 41.1 36.4 41.6
[61] 35.5 41.1 35.9 41.8 33.5 39.7 39.6 45.8 35.5 42.8 40.9 37.2 36.2 42.1 34.6
[76] 42.9 36.7 35.1 37.3 41.3 36.3 36.9 38.3 38.9 35.7 41.1 34.0 39.6 36.2 40.8
[91] 38.1 40.3 33.1 43.2 35.0 41.0 37.7 37.8 37.9 39.7 38.6 38.2 38.1 43.2 38.1
[106] 45.6 39.7 42.2 39.6 42.7 38.6 37.3 35.7 41.1 36.2 37.7 40.2 41.4 35.2 40.6
```

```
[121] 38.8 41.5 39.0 44.1 38.5 43.1 36.8 37.5 38.1 41.1 35.6 40.2 37.0 39.7 40.2  
[136] 40.6 32.1 40.7 37.3 39.0 39.2 36.6 36.0 37.8 36.0 41.5
```

```
length(adelie_bills)
```

```
[1] 146
```

Number of Adelie penguins with complete data: 146.

3 (5 points)

Calculate the following descriptive statistics for Adelie penguin bill lengths: - Mean - Standard deviation

- Median - 25th and 75th percentiles

```
summary(adelie_bills)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
32.10	36.73	38.85	38.82	40.77	46.00

```
sd(adelie_bills)
```

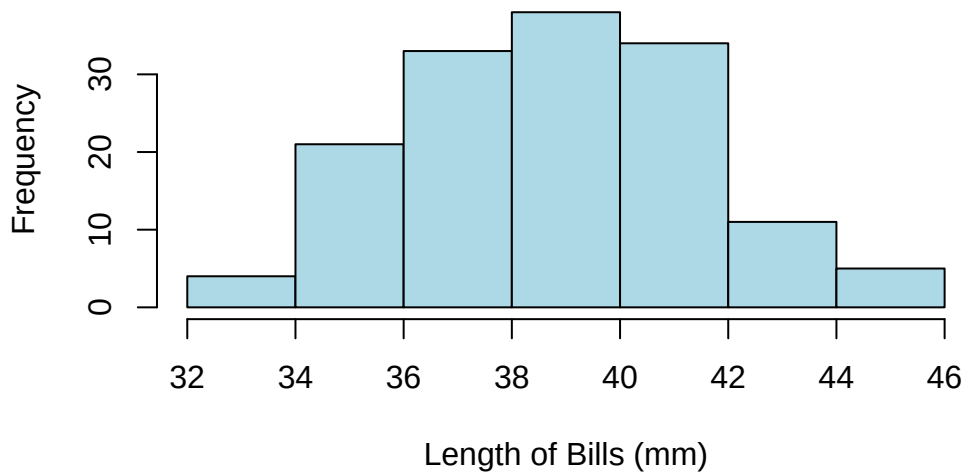
```
[1] 2.662597
```

4 (5 points)

Create a histogram of Adelie penguin bill lengths using ggplot2 or base R. Include appropriate axis labels and a title.

```
hist(adelie_bills, xlab="Length of Bills (mm)", main="Histogram of Bill Lengths for Adelie P")
```

Histogram of Bill Lengths for Adelie Penguins



Based on the histogram, briefly describe the distribution of bill lengths (address shape, center, and spread):

Description: The distribution seems normal and symmetric with very little outliers and one peak in the middle. The median bill length is 38.85 mm and the mean bill length is 38.82 mm, indicating very little skew. The standard deviation is 2.66. The middle 50% of the data is between 36.73 mm and 40.77 mm, with the shortest bill being 32.10 mm and the longest being 46 mm.

Statistical Analysis

1 (5 points)

Now we will formally test whether the population mean bill length μ differs from 39 inches, i.e. we will test:

$$H_0 : \mu = 39$$

against

$$H_A : \mu \neq 39$$

Perform a two-sided t-test to test these hypotheses. Use $\alpha = 0.05$.

```
t.test(x=adelie_bills,  
       mu=39,  
       alternative="two.sided",  
       alpha=0.05)
```

One Sample t-test

```
data: adelie_bills  
t = -0.79882, df = 145, p-value = 0.4257  
alternative hypothesis: true mean is not equal to 39  
95 percent confidence interval:  
 38.38844 39.25950  
sample estimates:  
mean of x  
 38.82397
```

Based on your test results, what is your conclusion? Interpret the results in the context of the research question.

Conclusion: The p-value is 0.42, which is not smaller than $\alpha = 0.05$; we do not reject the null hypothesis. There is insufficient evidence that the mean bill length for Adelie penguins differ significantly from 39 mm.

2 (4 points)

Define another variable (a numeric vector) `confidence_interval` and assign as its value the two bounds (lower and upper) of a 95% confidence interval for the mean bill length of Adelie penguins. Print the result. Full points are possible only if you do not “hard code” the bounds for the interval, i.e. if you create this vector directly by using R functions applied to the dataset. Half points maximum if you define the vector by typing the entries manually.

```
confidence_interval = t.test(x=adelie_bills,  
                             level=0.95)$conf.int  
confidence_interval
```

```
[1] 38.38844 39.25950  
attr(,"conf.level")  
[1] 0.95
```
